MATH503 Mathematics for Computing

Tutorial for Week 8

Question 1

A principal claim that the students in his school are of above average intelligence. A random sample of 30 IQ scores have a mean score of 112.5. Is there sufficient evidence to support the principal's claim? Use a 5 % significance level to test the principal's claim. The mean population IQ is 100 with a standard deviation of 15.

Question 2

A neurologist is testing the effect of a drug on response time by injecting 100 rats with a unit dose of the drug, subjecting each to neurological stimulus, and recording its response time. The neurologist knows that the mean response time for rats not injected with the drug is 1.2 seconds. The mean of the 100 injected rats' response time is 1.05 seconds with a sample standard deviation of 0.5 seconds. Do you think that the drug has reduced the response time? Use a 5% significance level to test if the drug has an effect.

Note: The standard deviation of the population is unknown. In this case we should use a t-test (we haven't covered t-tests in this course), **however**, because the sample size is greater than 30, so we can use the normal distribution.

Question 3

In Luigi's restaurant, on average 1 in 10 people order a bottle of Chardonnay. Out of a sample of 50 people, 11 chose Chardonnay. Has the drink become more popular? Test at the 1% level of significance.

Question 4

A person suggests that the proportion, p of red cars on a road is 0.3. In a random sample of 15 cars it is desired to test the null hypothesis $p = 0.3$ against the alternative hypothesis $p \neq 0.3$ at a significance level of 10%. Determine the appropriate region and the appropriate rejection region.